

Guilherme Ribeiro Figueira

C++/C# Generalist Game Programmer / Engineer

🔗 github.com/grfigueira | [in linkedin.com/in/grfigueira](https://www.linkedin.com/in/grfigueira) | 🌐 My Portfolio

☎ (+351) 916099199 | ✉ gr.figueira@proton.me

Education

Nova School Of Science and Technology

Almada, PT

B.Sc. + M.Sc. in Computer Science and Engineering

2020 - 2025

- **Master Thesis:** *Dynamic Substitutional Reality In Adaptive Virtual Experiences*. Contributed to an **HCI** research gap by researching and developing a dynamic system for the Unity engine that enables the creation of SR experiences in an abstract manner, that can be deployed into arbitrary spaces.
- Co-authored an additional paper [P1](#).
- Part of my research and work is being ported to a **real museum application** the *Portuguese National Tile Museum*, in Lisbon, Portugal.
- **Relevant courses:** Games and Simulations, Computer Graphics, OOP, AI, Concurrent Programming

Publications

P1 Towards Augmented XR Movement and Interaction Confidence across Substitutional Reality

Guilherme R. Figueira, Rui Nóbrega

International Conference on Graphics and Interaction (ICGI), 2025

Experience

Unity XR Developer - Curricular Internship

03/2023 - 08/2023

Glartek

Leiria, PT (Hybrid)

- Implemented an AR 3D model manipulation feature for **Microsoft Hololens 2** using Unity and MRTK while collaborating with the web development and design teams, adding to the project's feature suit.
- Utilized **3D vector math**, **memoization techniques**, and **search algorithms** to optimize performance and ensure reliable processing of imported 3D model for the supported formats.
- Created **unit and integration tests** for scripts using **Zenject dependency injection** framework, improving test coverage and code reliability.
- Collaborated using an **AGILE** methodology with **GitLab** for version control.

Skills

Programming Languages: C++, C#, Java, Python, OCaml, HLSL

Engines & Frameworks: Unreal Engine 5, Unity, FMOD, ImGui, SDL2

Graphics & Tools: OpenGL, WebGL, Git, Shell, VIM, Docker

Creative Software: Blender (Basics), Photoshop, DaVinci Resolve

Languages: Portuguese (native), English (proficient)

Projects

The Shape of Silence - Unreal Engine 5 C++ Game

01/2024 - Present

Personal Game

- A narrative first person puzzle I am started making to sharpen my Unreal Engine C++ skills. I am working on this project solo which means doing all the game design, writing, sound design, and programming.
- So far I have developed an **extendable grid-based puzzle mechanic**, an **inventory subsystem** with support for item inspections, and a **dialogue system** using my own **EasyDialogueEditor** tool to write the dialogue.

Easy Dialogue Editor

03/2025 (3 weeks)

Personal Tool

- A FOSS, lightweight and platform agnostic **desktop program for editing game dialogue trees**. Created to solve a problem in a personal UE5 project.
- Written in **C++** with **ImGui**. It compiles to a **single small and portable executable** (~2.7 Mb) with no external dependencies, everything is statically linked.

OpenGL C++ 3D renderer

07/2025 - Present

Personal learning exercise

- Used C++ and OpenGL to implement a decoupled renderer, separating the rendering from the application and shaders.
- Currently implemented features include: **3D fly-style camera**, **textured 3D objects**, a **fog setting**, and a settings panel added with **ImGui** to control camera, window and fog settings.
- On-going project, with the goal of eventually growing into my own small 3D game engine

The Natural's Descent - Unity Game

04/2024 - 06/2024

University Games and Simulations Course

- Action-adventure puzzle game developed in Unity3D as part of a **2-person team**.
- Designed and implemented a **procedurally generated cavern** using the **Wave Function Collapse** algorithm adapted to 3D, ensuring natural looking layouts with procedurally spawned enemies.
- Programmed multiple core gameplay mechanics such as: a *extendable sigil puzzle** component, a **ridable vehicle** and **FPS combat with AI animated humanoid enemies**.